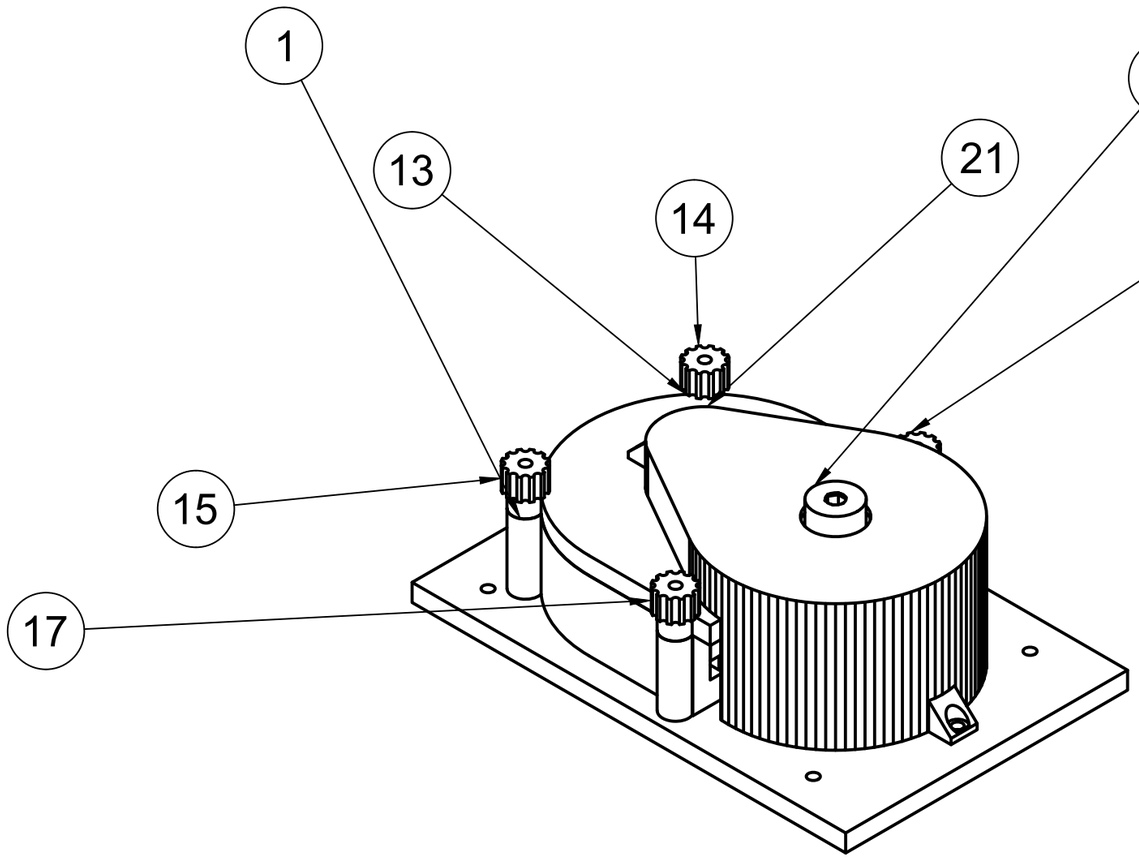
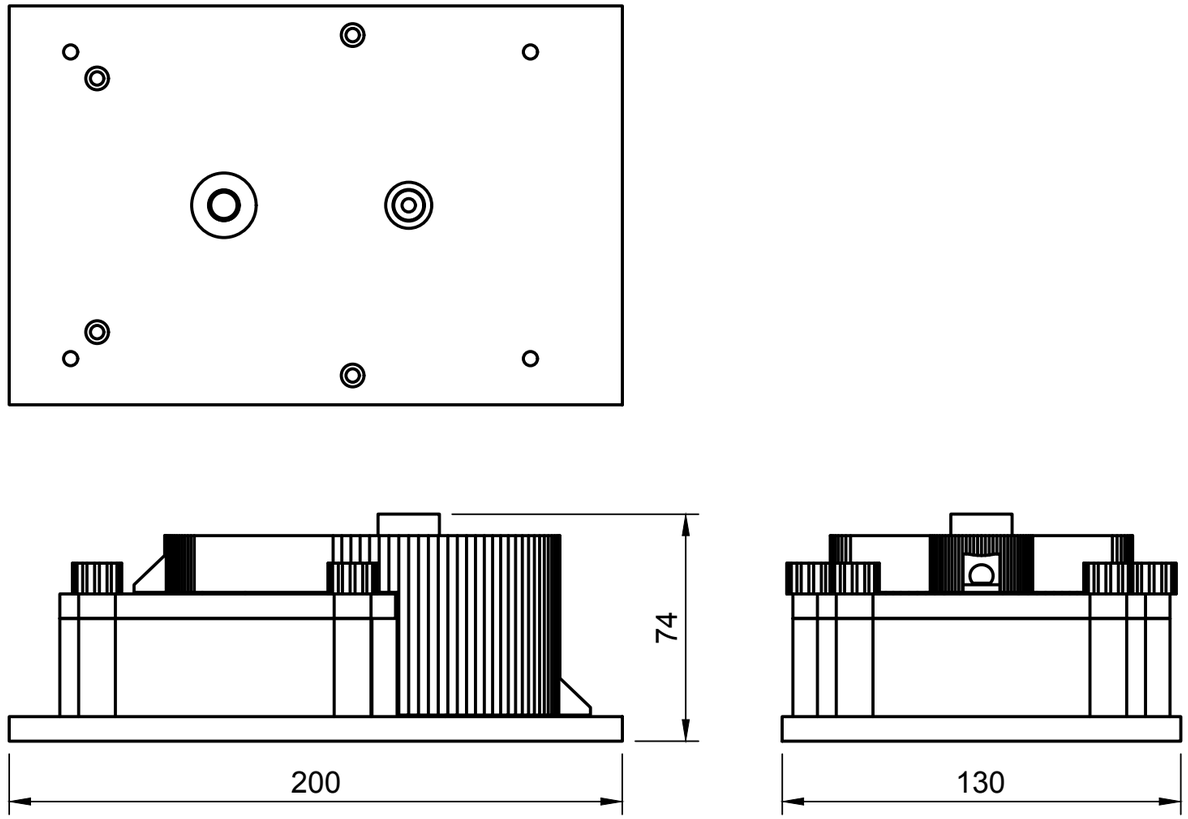


Parts list

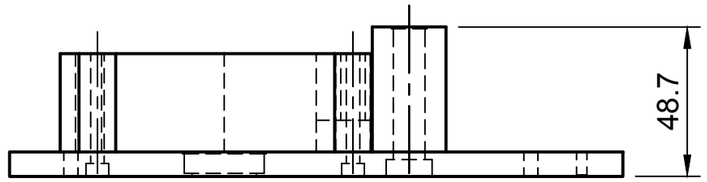
Item	Qty	Part number / description	Material
1	1	HB-01 base plate + housing	PLA, FDM
2	1	HB-03 race liner, gap 1.70 (also 1.50 / 1.90)	PLA, FDM
3	1	HB-04 rotor lower spider + shaft	PLA, FDM
4	1	HB-05 rotor upper spider	PLA, FDM
5	1	Ball bearing 6000 10x26x8, roller A lower	in hand, POM race
6	1	Ball bearing 6000 10x26x8, roller A upper	in hand, POM race
7	1	Ball bearing 6000 10x26x8, roller B lower	in hand, POM race
8	1	Ball bearing 6000 10x26x8, roller B upper	in hand, POM race
9	1	Ball bearing 6000 10x26x8, roller C lower	in hand, POM race
10	1	Ball bearing 6000 10x26x8, roller C upper	in hand, POM race
11	1	Ball bearing 6000 10x26x8, rotor lower	in hand, POM race
12	1	Ball bearing 6000 10x26x8, rotor upper	in hand, POM race
13	1	HB-02 lid	PLA, FDM
14	1	HB-09 thumb nut A (+ M4 nut)	PLA, FDM
15	1	HB-09 thumb nut B (+ M4 nut)	PLA, FDM
16	1	HB-09 thumb nut C (+ M4 nut)	PLA, FDM
17	1	HB-09 thumb nut D (+ M4 nut)	PLA, FDM
18	1	HB-06 pinion 20 T, m1.5	PLA, FDM
19	1	HB-07 crank wheel 60 T + hex socket	PLA, FDM
20	1	HB-08 crank shaft	PLA, FDM
21	1	HB-13 gear cover	PLA, FDM

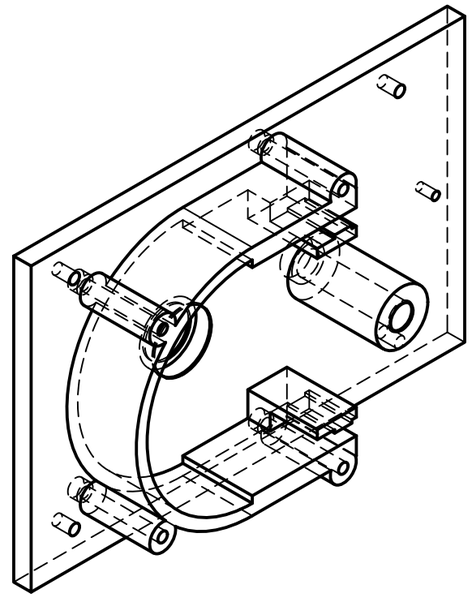
Bearings: the 8 in hand (9 Nov photo) have white POM races, a POM cage and steel balls. Checked OK for this design: race flexes about 0.04 mm under a 40 N roller load (far below the 0.2 mm liner steps), no rust in water, same 10x26x8 envelope. Measure OD, ID and width before printing and set them in params_home.py. Those 8 are the TEAM kit's set: for the home build buy your own 8x 6000-2RS so the team keeps its parts. Hardware (not shown): 3x M4x25 + nut (rotor posts); 4x M4x50 (lid, into the thumb nuts); 1x M4x60 + dia 12 penny washer + nut (crank); 1x M4x12 + washer (pinion); 2x M4x10 self-tapping (gear cover). Kit screws are Phillips pan head: open the counterbores to dia 8.6 if you use them. Silicone tube 5x7, about 0.6 m. Drive: 6 mm A/F hex key.

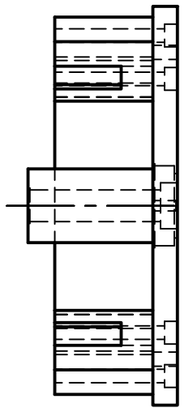
Dept.	Technical reference	Created by	Approved by
		paing htet 09/11/2026	
		Document type: General assembly	Document status: home build v2, AI-authored CAD, not for submission
		Title: pump_home_v2_drawing_src	DWG No.: HB-00 GA
			Rev. / Date of issue / Sheet: 1/11

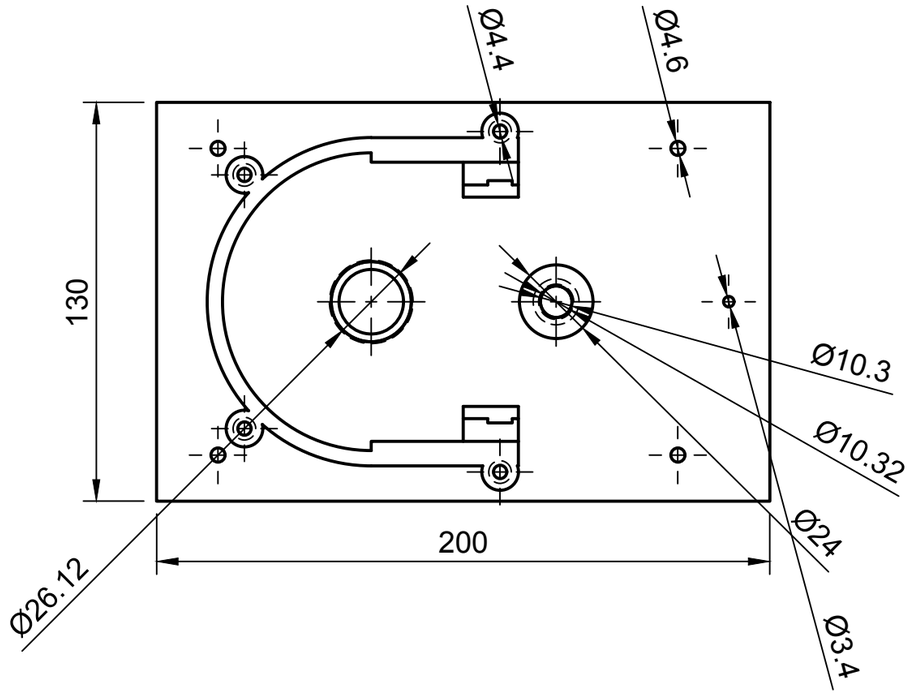


Dept.	Technical reference	Created by	Approved by
		paing htet 09/11/2026	
		Document type	Document status
		Title: pump_home_v2_drawing_src	DWG No.
			Rev. / Date of issue / Sheet: 2/11

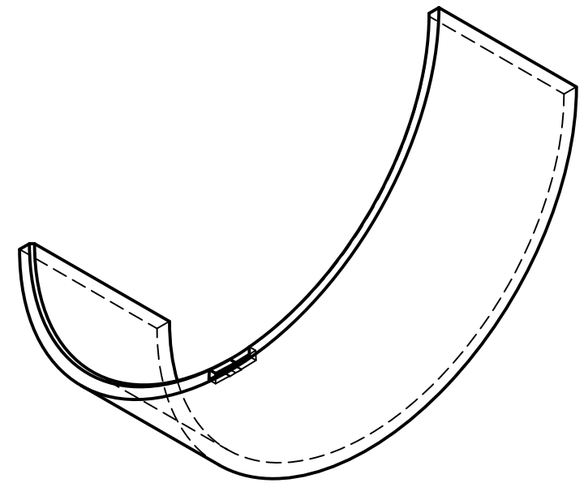


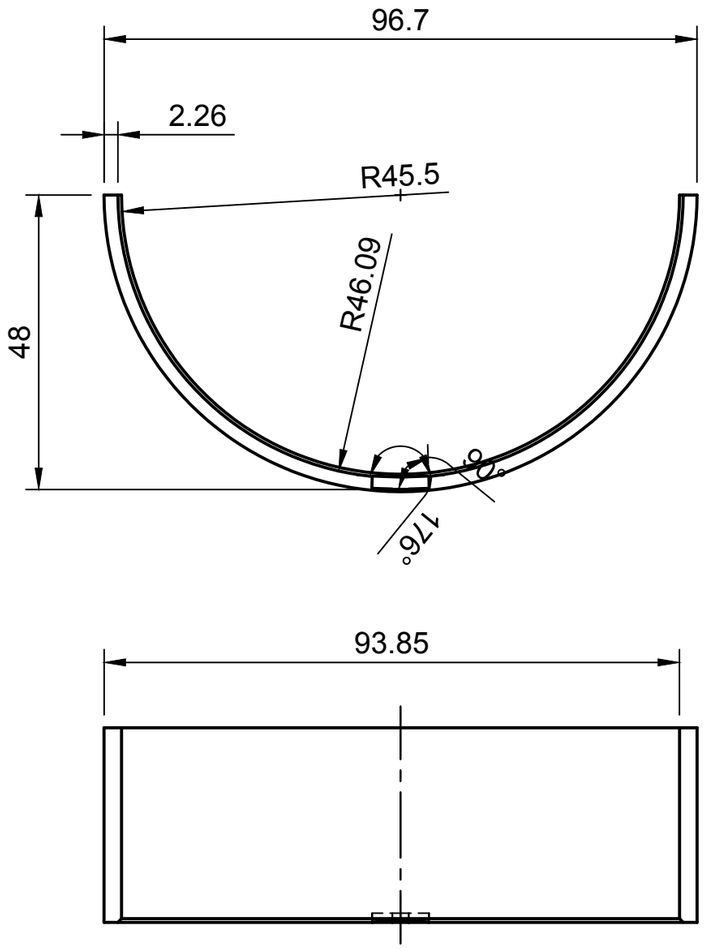


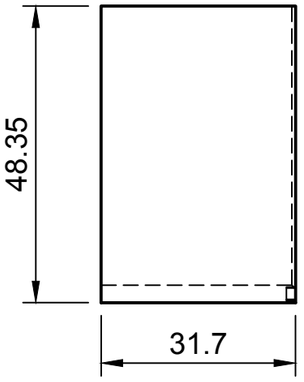




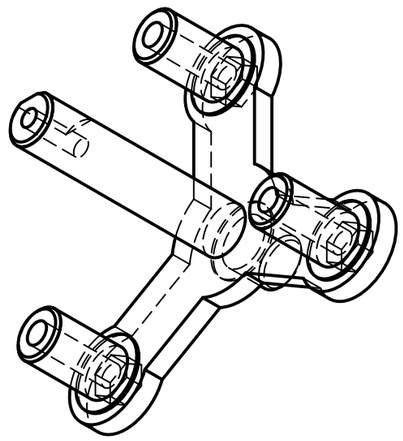
Dept.	Technical reference	Created by paing htet 09/11/2026	Approved by
		Document type	Document status
		Title: pump_home_v2_drawing_src	DWG No.
			Rev. / Date of issue / Sheet 3/11

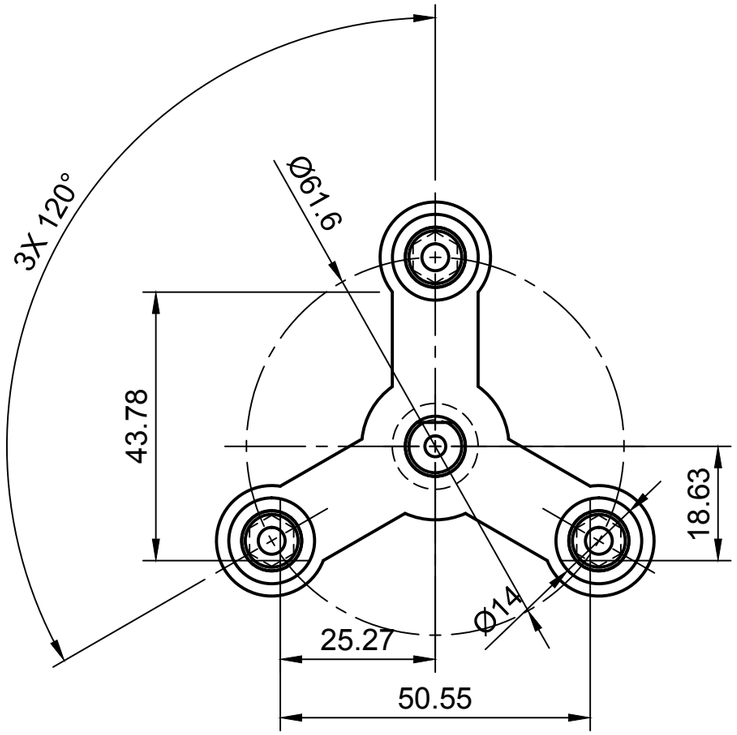


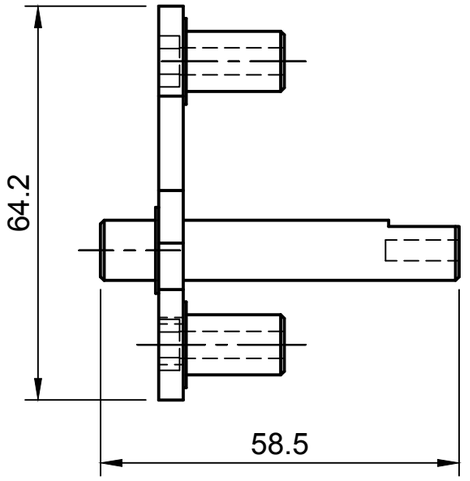


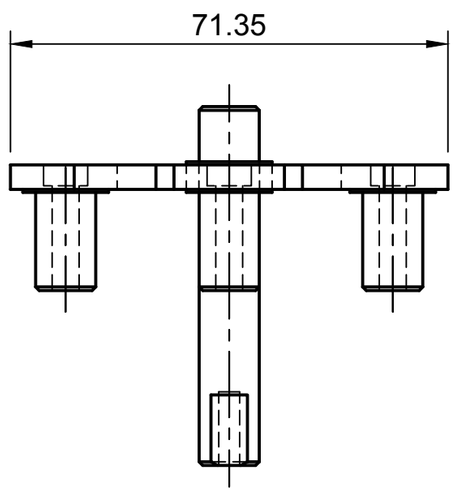


Dept.	Technical reference	Created by paing htet 09/11/2026	Approved by
		Document type	Document status
		Title pump_home_v2_drawing_src	DWG No.
		Rev. / Date of issue	Sheet 4/11

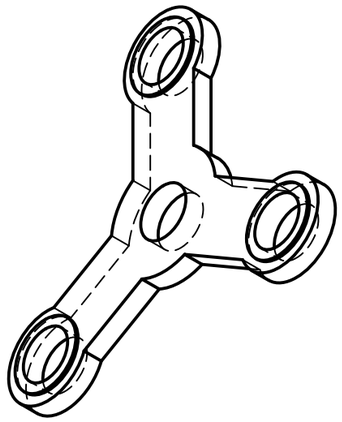


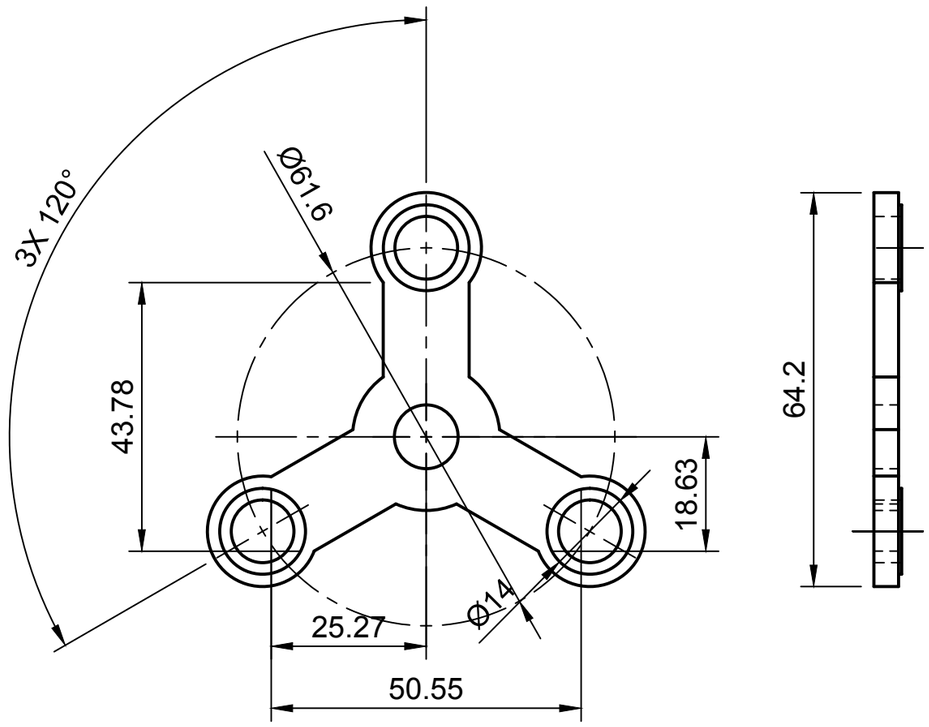


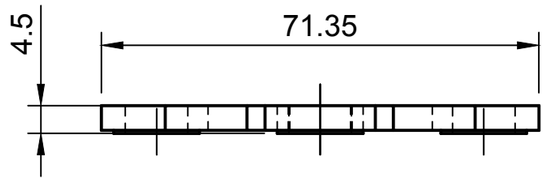




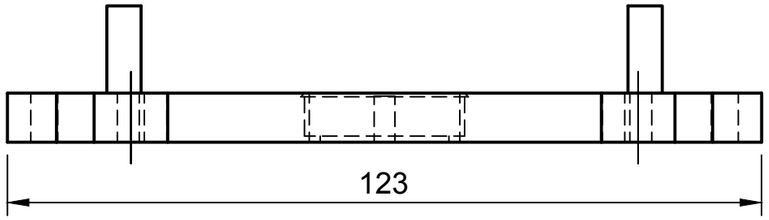
Dept.	Technical reference	Created by paing htet 09/11/2026	Approved by
		Document type	Document status
		Title pump_home_v2_drawing_src	DWG No.
			Rev. / Date of issue / Sheet 5/11

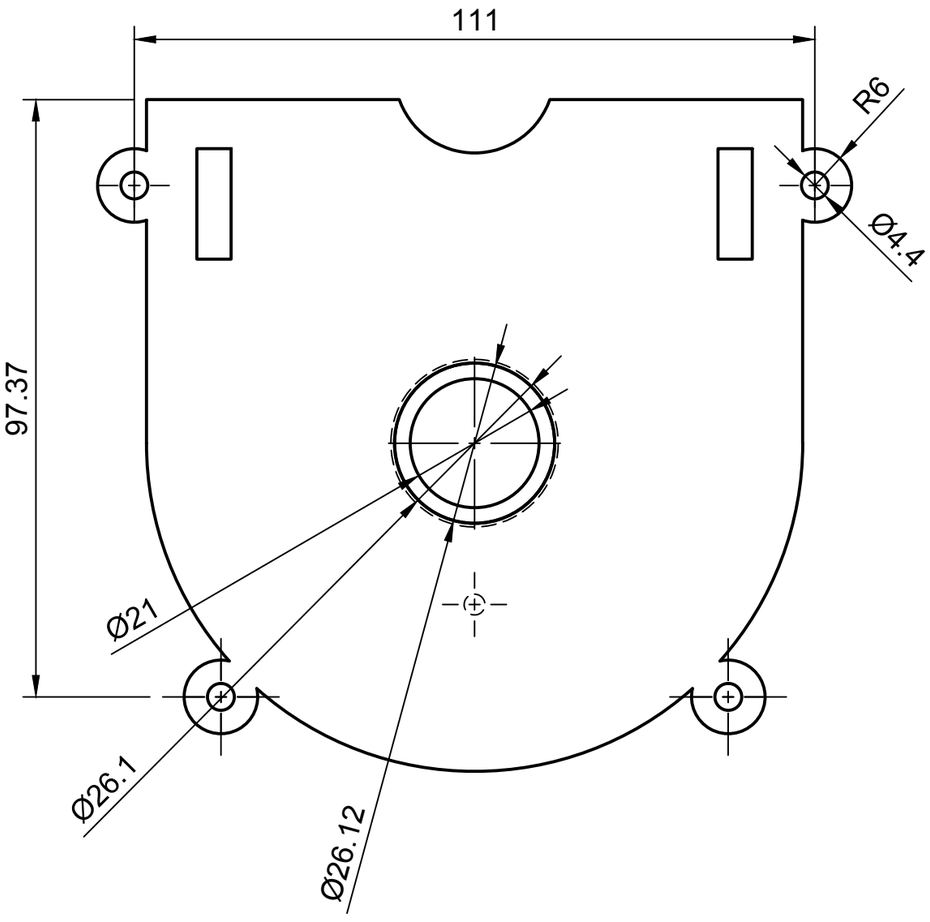


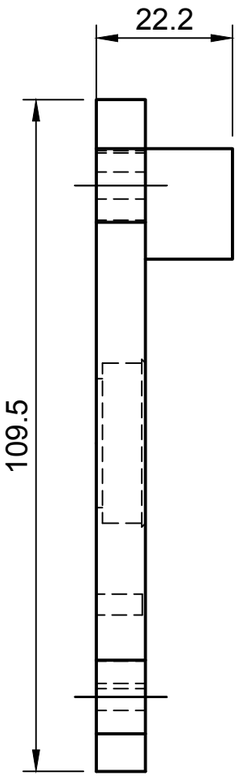


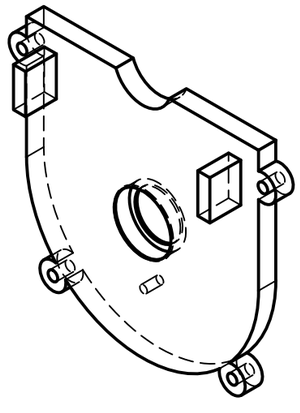


Dept.	Technical reference	Created by: paing htet 09/11/2026	Approved by
		Document type	Document status
		Title: pump_home_v2_drawing_src	DWG No.
			Rev. / Date of issue / Sheet: 6/11





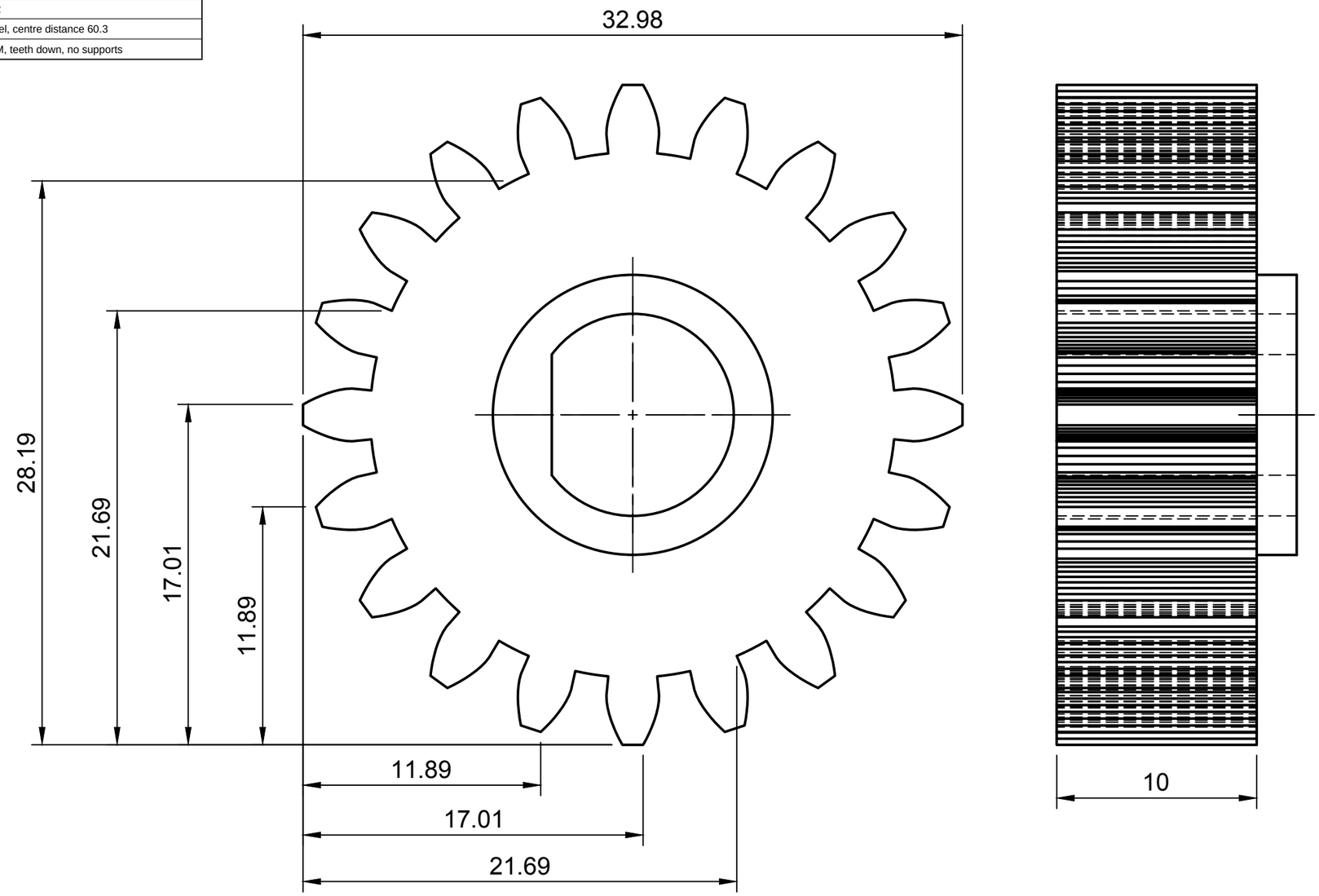


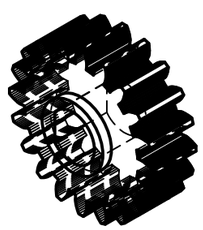


Dept.	Technical reference	Created by paing htet 09/11/2026	Approved by
		Document type	Document status
		Title pump_home_v2_drawing_src	DWG No.
			Rev. / Date of issue / Sheet 7/11

GEAR DATA: PINION (sheet 8)

Parameter	Value
Teeth	20
Module	1.5
Pressure angle	20 deg
Pitch diameter	30.00
Outside diameter	33.00
Root diameter	26.25
Face width	10
Bore	dia 10.1, D flat 4.05 from the axis
Thrust boss	dia 14 x 2
Mating gear	60 T wheel, centre distance 60.3
Material / process	PLA, FDM, teeth down, no supports

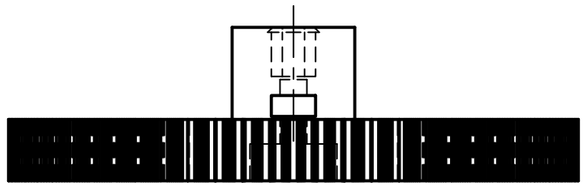


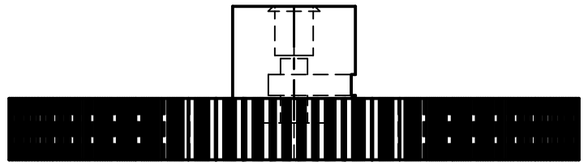


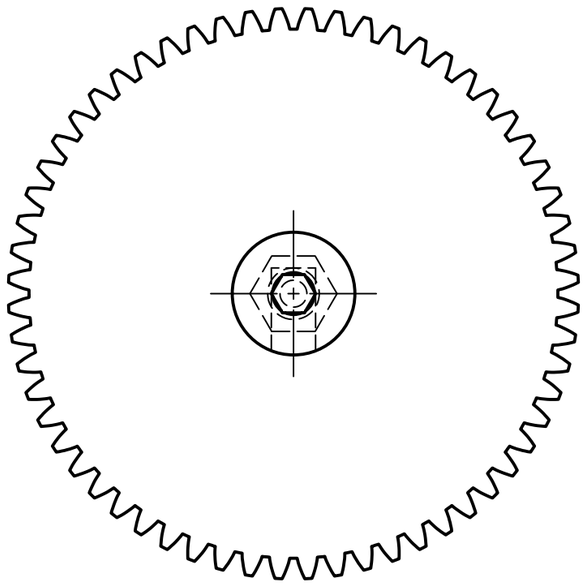
Dept.	Technical reference	Created by	Approved by
		paing htet 09/11/2026	
		Document type	Document status
		Title: pump_home_v2_drawing_src	DWG No.
			Rev. / Date of issue / Sheet: 8/11

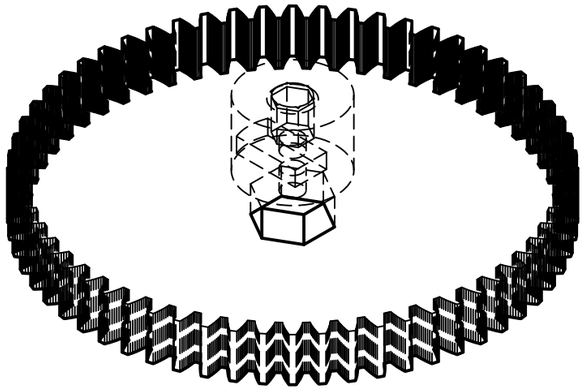
GEAR DATA: CRANK WHEEL (sheet 9)

Parameter	Value
Teeth	60
Module	1.5
Pressure angle	20 deg
Pitch diameter	90.00
Outside diameter	93.00
Root diameter	86.25
Face width	10
Hub boss	dia 20 x 15 above the gear face
Hex socket	6.25 A/F x 8 deep, 0.6 chamfer
Shaft hex pocket (underside)	12.3 A/F x 6 deep
Bolt bore / nut slot	dia 4.4; M4 nut slot 7.2 x 3.4, 10.5 above the gear face
Mating gear	20 T pinion, centre distance 60.3, ratio 3:1
Material / process	PLA, FDM, gear face down, no supports

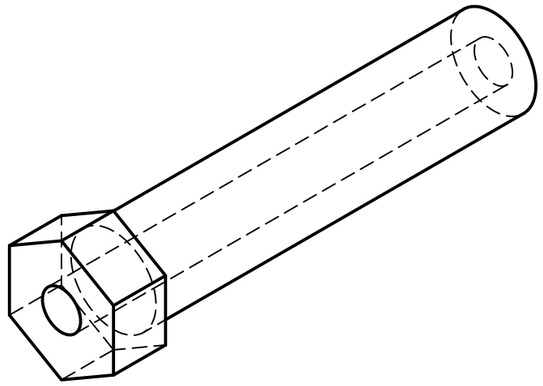


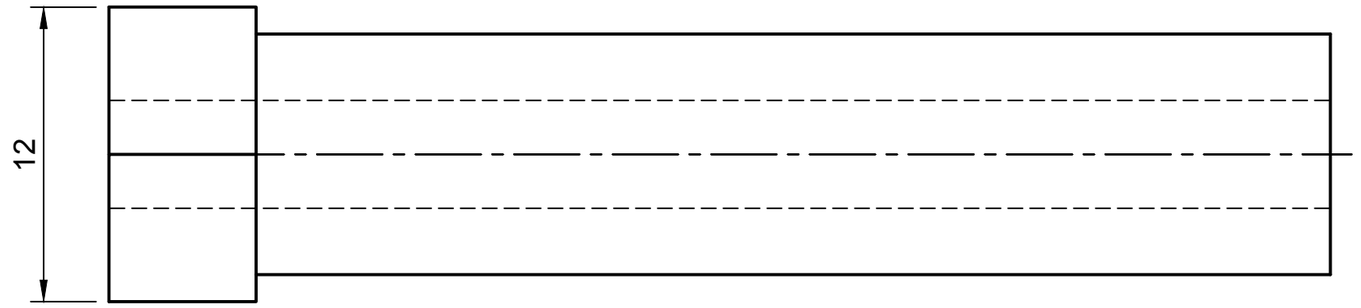


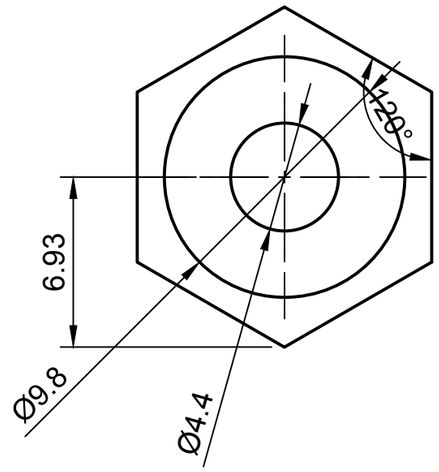


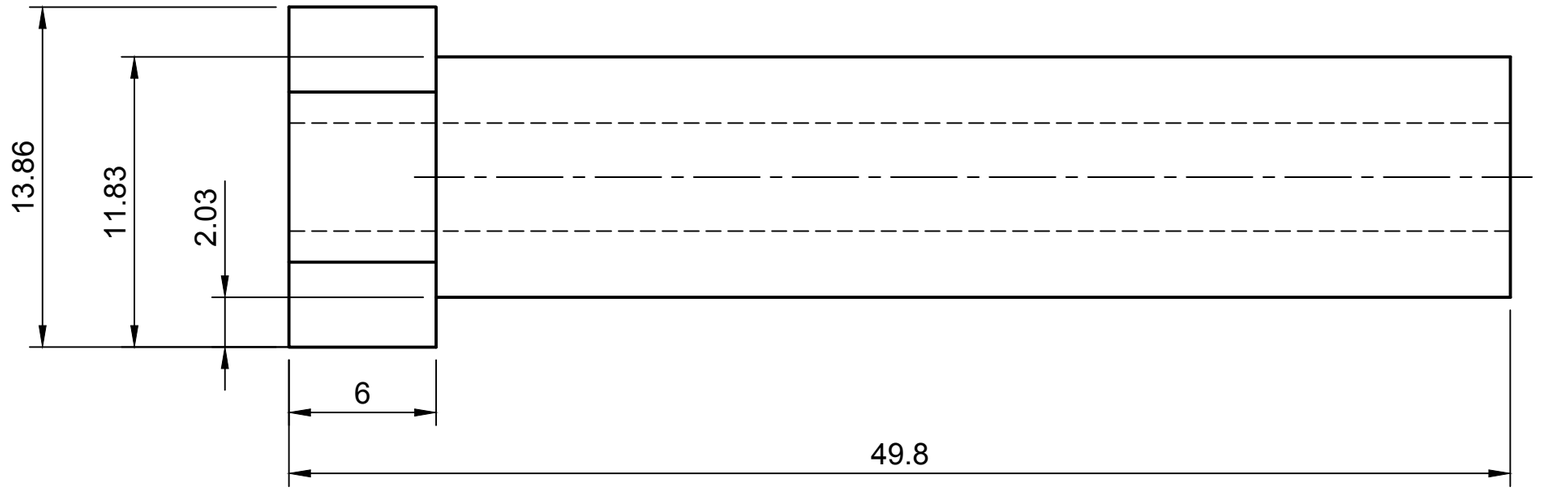


Dept.	Technical reference	Created by	Approved by
		paing htet 09/11/2026	
		Document type	Document status
		Title: pump_home_v2_drawing_src	DWG No.
			Rev. / Date of issue / Sheet: 9/11

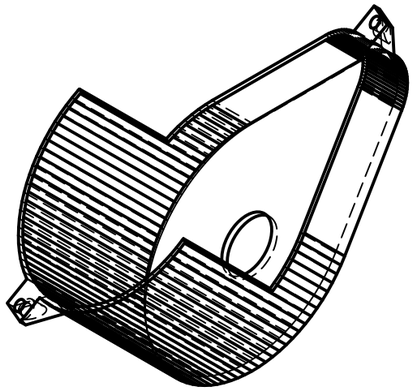


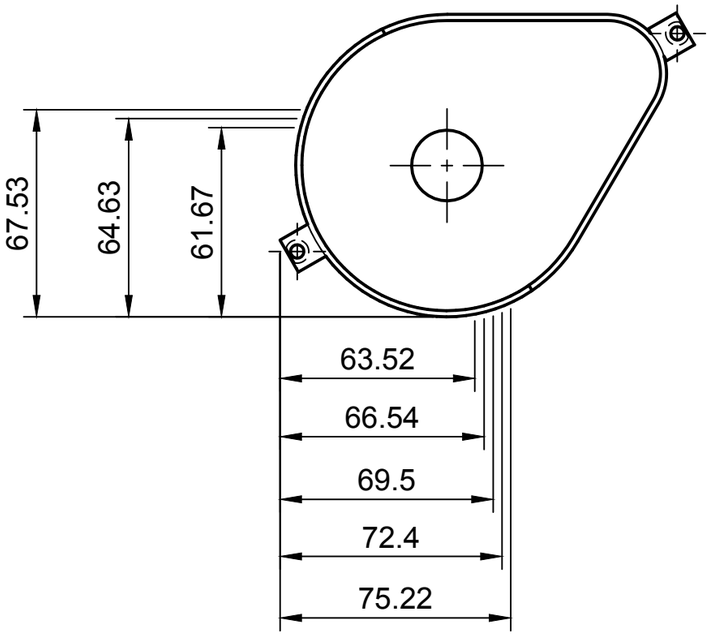


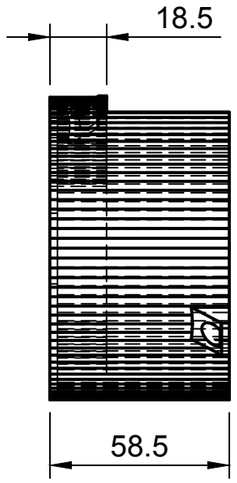


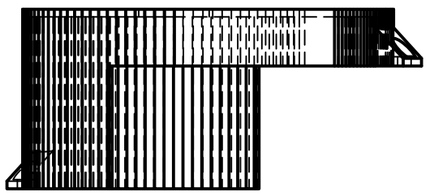


Dept.	Technical reference	Created by paing htet 09/11/2026	Approved by
		Document type	Document status
		Title: pump_home_v2_drawing_src	DWG No.
			Rev. / Date of issue / Sheet: 10/11









Dept.	Technical reference	Created by paing htet 09/11/2026	Approved by
		Document type	Document status
		Title: pump_home_v2_drawing_src	DWG No.
			Rev. / Date of issue / Sheet: 11/11